

# Horizontal and vertical dilations



1

a 360

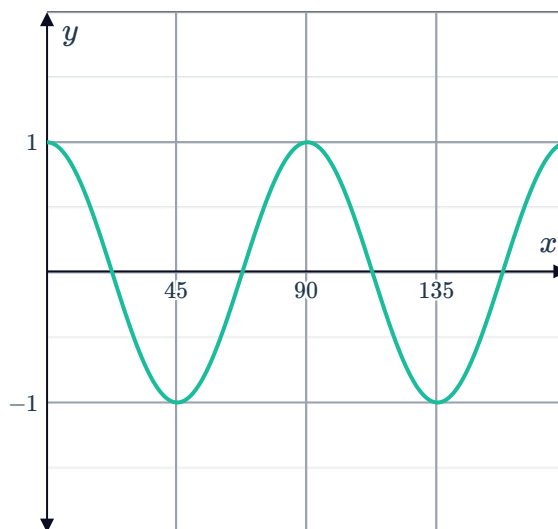
b

|        |   |      |    |      |    |       |     |       |     |
|--------|---|------|----|------|----|-------|-----|-------|-----|
| $x$    | 0 | 22.5 | 45 | 67.5 | 90 | 112.5 | 135 | 157.5 | 180 |
| $g(x)$ | 1 | 0    | -1 | 0    | 1  | 0     | -1  | 0     | 1   |

c 90

d Horizontal dilation by a factor of  $\frac{1}{4}$

e

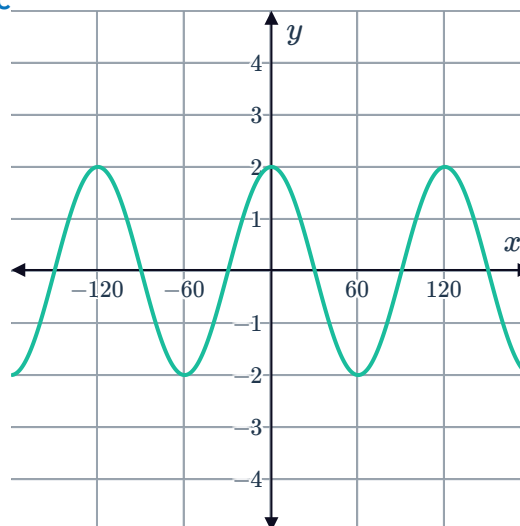


2

a 2

b 120

c



3

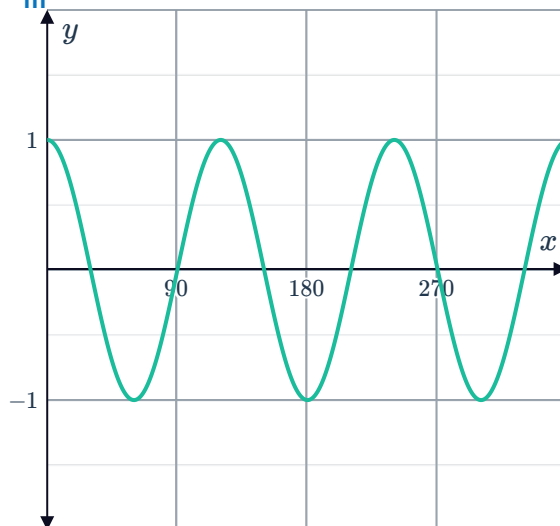
a

i  $A = 1$

ii  $P = 120$



iii

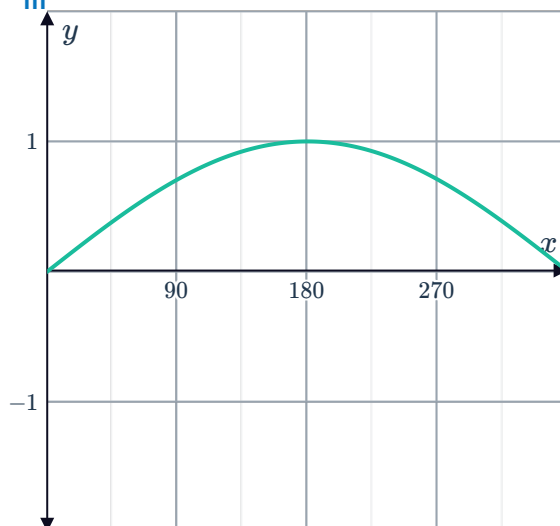


b

i  $A = 1$

ii  $P = 720$

iii

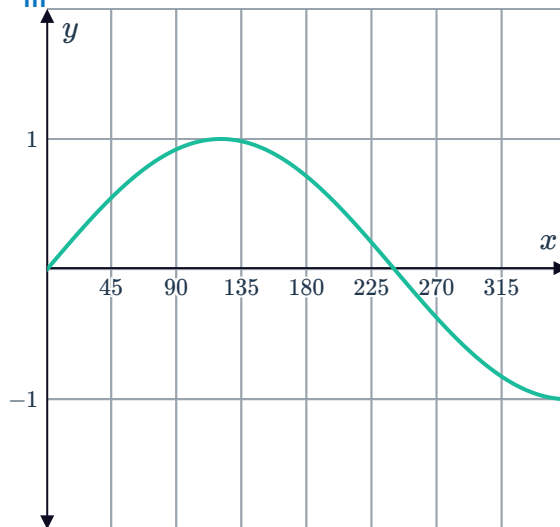


c

i  $A = 1$

ii  $P = 480$

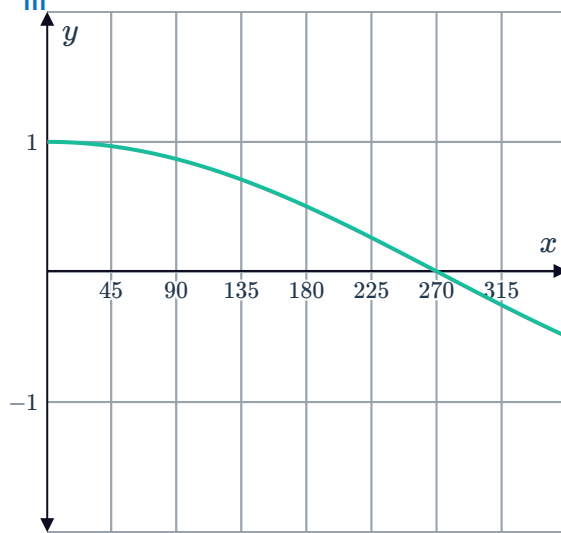
iii



d

i  $A = 1$

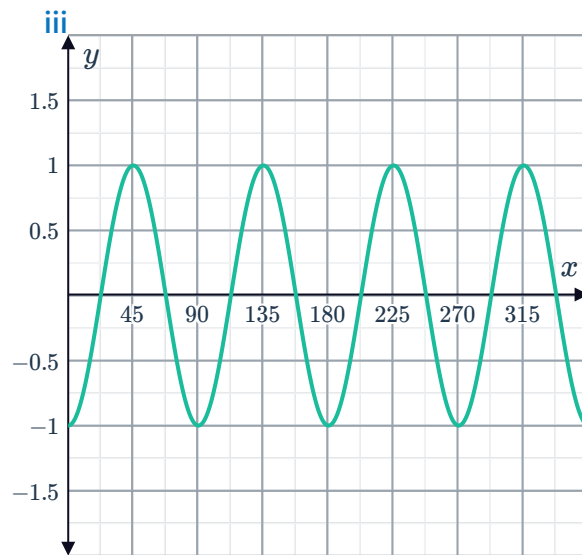
ii  $P = 1080$



e

i  $A = 1$

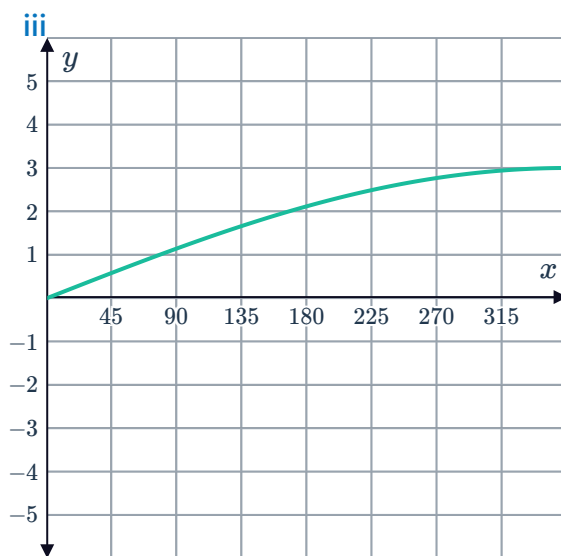
ii  $P = 90$



f

i  $A = 3$

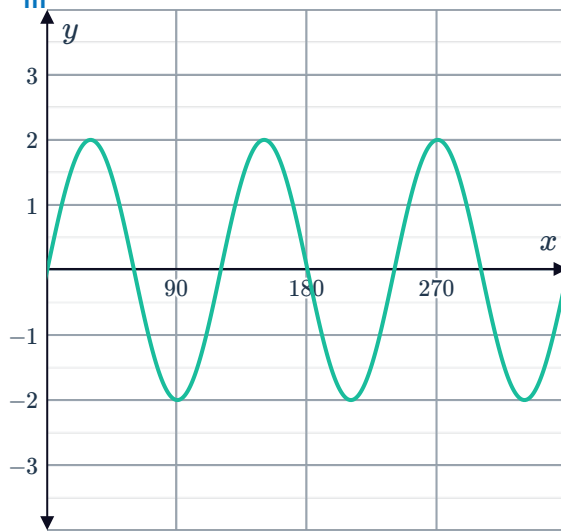
ii  $P = 1440$



g

i  $A = 2$

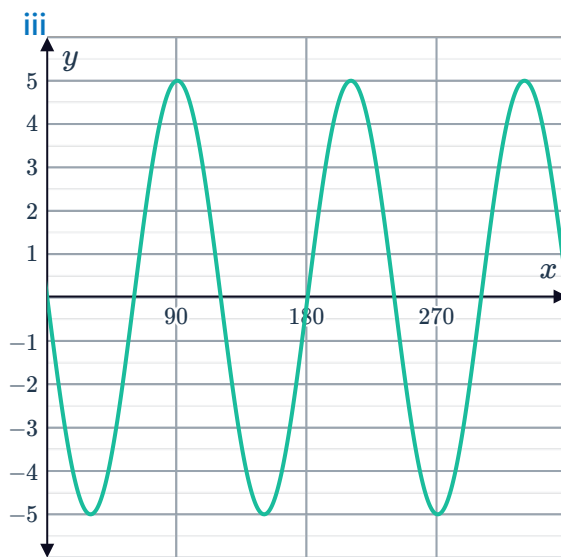
ii  $P = 120$



h

i  $A = 5$

ii  $P = 120$

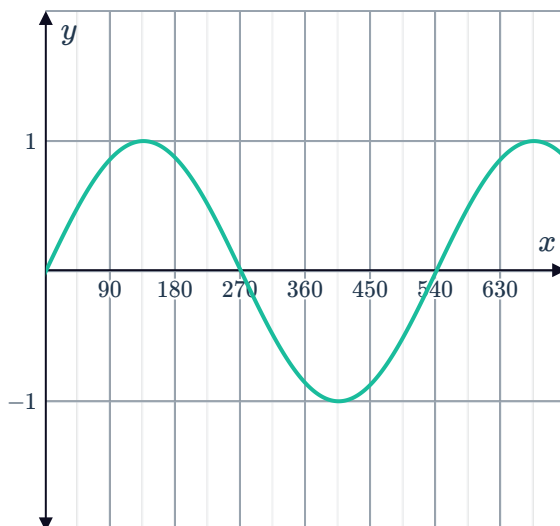


4

a 1

b  $540^\circ$

c



5



a

|                                |   |     |     |      |      |
|--------------------------------|---|-----|-----|------|------|
| $x$                            | 0 | 360 | 720 | 1080 | 1440 |
| $\sin\left(\frac{x}{4}\right)$ | 0 | 1   | 0   | -1   | 0    |

b 1440

6

a Horizontal dilation by a factor of  $\frac{1}{4}$ .

b 4

### Combined translations and dilations

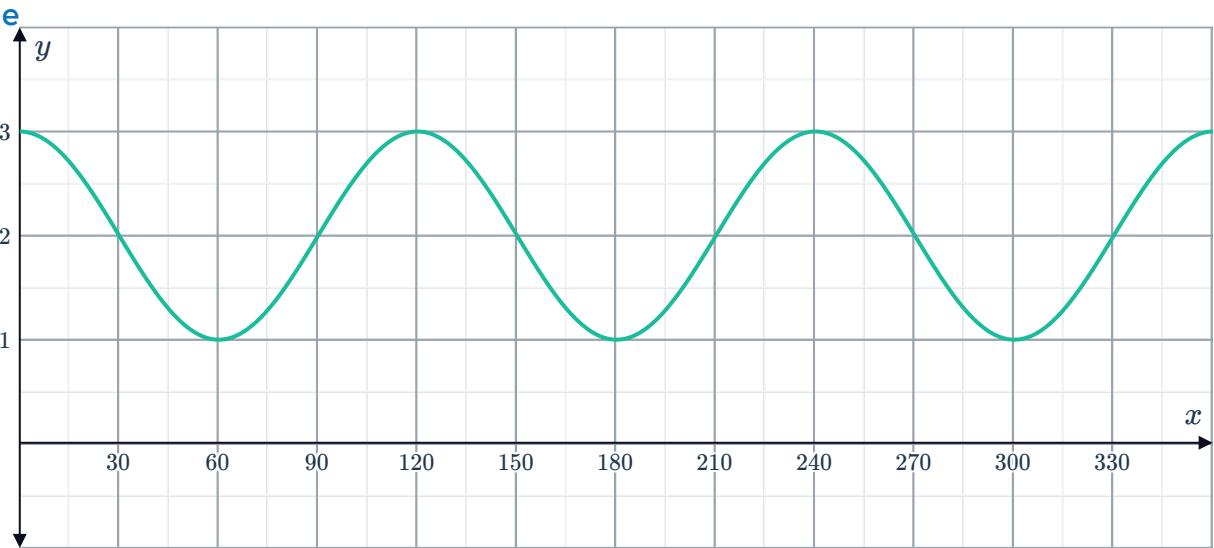
7

a 120

b 1

c 3

d 1



8

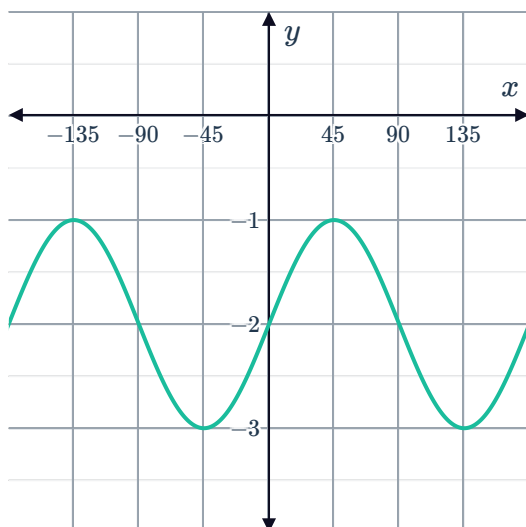


a

i  $(-\infty, \infty)$

ii  $[-3, -1]$

iii

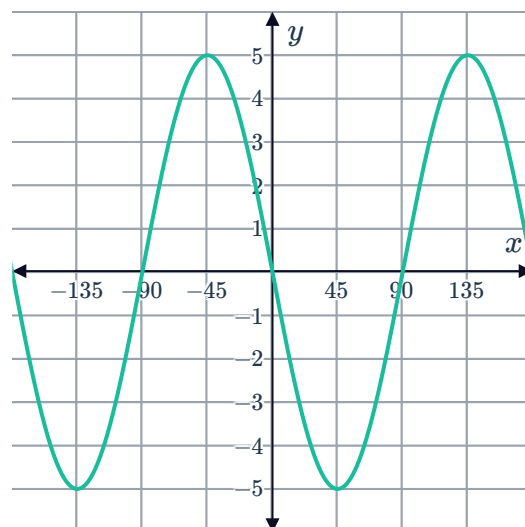


b

i  $(-\infty, \infty)$

ii  $[-5, 5]$

iii

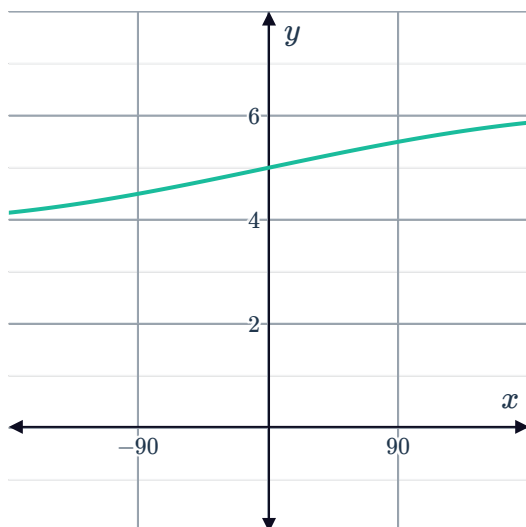


c

i  $(-\infty, \infty)$

ii  $[4, 6]$

iii

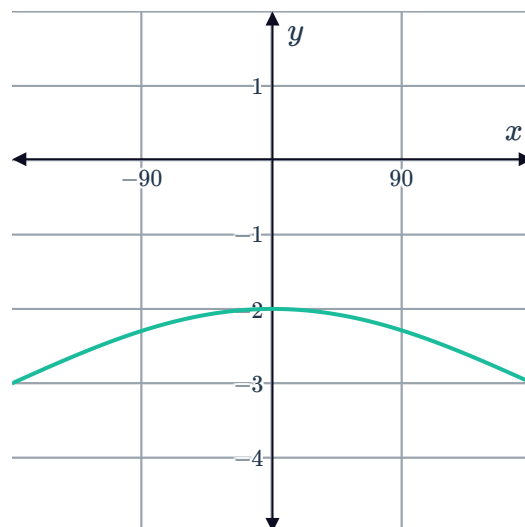


d

i  $(-\infty, \infty)$

ii  $[-4, -2]$

iii



9

a  $y = 3 \sin 2x$

b  $y = -\sin 2x$

c  $y = 2 \cos 3x$

d  $y = -4 \cos \left( \frac{x}{2} \right)$

10

a Yes

b No

c Yes

d No